

Supplementary Material for: “ELACS-Net: From Elastic Regularization to Deep Elastic Network for Compressed Sensing”

A. PROOF OF THEOREM 1

Proof. We provide a detailed proof of Theorem 1, which establishes the relationship between the expected reconstruction error and the expected measurement error under the assumption of independent and standardized input variables.

A.1 Problem Setup and Assumptions

Assumptions:

- The inputs $x_1, x_2, \dots, x_n$ are independent and standardized random variables, meaning:
 - 1) $E[x_i] = 0$ for all $i = 1, 2, \dots, n$ (zero mean),
 - 2) $\text{Var}(x_i) = E[x_i^2] = 1$ for all $i = 1, 2, \dots, n$ (unit variance).
- The sensing matrix $\Phi \in \mathbb{R}^{m \times n}$ is a deterministic (non-random) matrix.
- Let $x = [x_1, \dots, x_n]^\top$ be the signal vector.
- The observation is given by $y = \Phi x \in \mathbb{R}^m$.

Goal: Prove that:

$$E(\|x - E[x]\|_2^2) = c \cdot E(\|y - E[y]\|_2^2), \quad (\text{S.1})$$

where $c = \frac{n}{\text{trace}(\Phi^\top \Phi)}$ is a constant depending only on the sensing matrix Φ .

A.2 Preliminary Results

Lemma A.1: Under the assumption that $E[x_i] = 0$ for all i , we have:

$$E[x] = E \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} E[x_1] \\ E[x_2] \\ \vdots \\ E[x_n] \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0. \quad (\text{S.2})$$

Proof of Lemma A.1: This follows directly from the linearity of expectation and the assumption that each component has zero mean. $\square$

A.3 Expectation of the Observation

From the observation model $y = \Phi x$ and the fact that Φ is deterministic, we have:

$$\begin{aligned} E[y] &= E[\Phi x] \\ &= \Phi E[x] \quad (\text{linearity of expectation and deterministic } \Phi) \\ &= \Phi \cdot 0 \quad (\text{by Lemma A.1}) \\ &= 0. \end{aligned} \quad (\text{S.3})$$

Therefore, both the signal and observation have zero mean:

$$E[x] = 0 \quad \text{and} \quad E[y] = 0. \quad (\text{S.4})$$

A.4 Computing the Expected Signal Error

The expected squared error of the signal is:

$$\begin{aligned} E(\|x - E[x]\|_2^2) &= E(\|x - 0\|_2^2) \quad (\text{by equation (S.2)}) \\ &= E(\|x\|_2^2) \\ &= E\left(\sum_{i=1}^n x_i^2\right) \\ &= \sum_{i=1}^n E[x_i^2] \quad (\text{linearity of expectation}). \end{aligned} \quad (\text{S.5})$$

By the standardization assumption, each x_i has zero mean and unit variance. Using the relationship between variance and second moment:

$$\begin{aligned} \text{Var}(x_i) &= E[x_i^2] - (E[x_i])^2 \\ &\Rightarrow E[x_i^2] = \text{Var}(x_i) + (E[x_i])^2 = 1 + 0 = 1. \end{aligned} \quad (\text{S.6})$$

Therefore:

$$E(\|x - E[x]\|_2^2) = \sum_{i=1}^n E[x_i^2] = \sum_{i=1}^n 1 = n. \quad (\text{S.7})$$

A.5 Computing the Expected Measurement Error

The expected squared error of the measurements is:

$$\begin{aligned} E(\|y - E[y]\|_2^2) &= E(\|y - 0\|_2^2) \quad (\text{by equation (S.3)}) \\ &= E(\|y\|_2^2) \\ &= E(\|\Phi x\|_2^2) \\ &= E((\Phi x)^\top (\Phi x)) \\ &= E(x^\top \Phi^\top \Phi x). \end{aligned} \quad (\text{S.8})$$

Let $A = \Phi^\top \Phi \in \mathbb{R}^{n \times n}$. Since Φ is deterministic, A is also deterministic, and we can move it out of the expectation:

$$\begin{aligned} E(x^\top A x) &= E\left(\sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n A_{ij} E[x_i x_j] \quad (\text{linearity and deterministic } A). \end{aligned} \quad (\text{S.9})$$

Since $x_1, x_2, \dots, x_n$ are independent with zero mean, we have:

$$E[x_i x_j] = \begin{cases} E[x_i^2] = 1, & \text{if } i = j \\ E[x_i]E[x_j] = 0 \cdot 0 = 0, & \text{if } i \neq j \end{cases} \quad (\text{S.10})$$

The second case follows from independence: for $i \neq j$, $E[x_i x_j] = E[x_i]E[x_j]$ since the variables are independent.

Therefore:

$$\begin{aligned}
 E(x^\top A x) &= \sum_{i=1}^n \sum_{j=1}^n A_{ij} E[x_i x_j] \\
 &= \sum_{i=1}^n A_{ii} E[x_i^2] \quad (\text{only diagonal terms survive}) \\
 &= \sum_{i=1}^n A_{ii} \cdot 1 \\
 &= \sum_{i=1}^n A_{ii} \\
 &= \text{trace}(A) \\
 &= \text{trace}(\Phi^\top \Phi). \tag{S.11}
 \end{aligned}$$

Thus:

$$E(\|y - E[y]\|_2^2) = \text{trace}(\Phi^\top \Phi). \tag{S.12}$$

A.6 Establishing the Constant Relationship

From equations (S.7) and (S.12), we have:

$$\frac{E(\|x - E[x]\|_2^2)}{E(\|y - E[y]\|_2^2)} = \frac{n}{\text{trace}(\Phi^\top \Phi)}. \tag{S.13}$$

Define the constant:

$$c = \frac{n}{\text{trace}(\Phi^\top \Phi)}. \tag{S.14}$$

Then we obtain the main result:

$$E(\|x - E[x]\|_2^2) = c \cdot E(\|y - E[y]\|_2^2), \tag{S.15}$$

where c is a constant that depends only on the sensing matrix Φ through the explicit formula $c = \frac{n}{\text{trace}(\Phi^\top \Phi)}$.

A.7 Properties and Interpretation of the Constant c

Property 1 (Orthonormal sensing): For an orthonormal sensing matrix where $\Phi^\top \Phi = I_n$ (e.g., when $m = n$ and Φ is orthogonal), we have:

$$c = \frac{n}{\text{trace}(I_n)} = \frac{n}{n} = 1. \tag{S.16}$$

This means the expected errors are equal, which aligns with the isometry property of orthogonal transformations.

Property 2 (Random Gaussian sensing): For a sensing matrix Φ whose entries are i.i.d. with $\Phi_{ij} \sim \mathcal{N}(0, 1/m)$, we have:

$$\begin{aligned}
 E[\text{trace}(\Phi^\top \Phi)] &= E\left[\sum_{j=1}^n \sum_{i=1}^m \Phi_{ij}^2\right] \\
 &= \sum_{j=1}^n \sum_{i=1}^m E[\Phi_{ij}^2] \\
 &= mn \cdot \frac{1}{m} = n. \tag{S.17}
 \end{aligned}$$

Thus, for such random matrices, $E[c] = \frac{n}{E[\text{trace}(\Phi^\top \Phi)]} \approx 1$ on average.

Property 3 (Energy preservation): The constant c can be interpreted as an energy scaling factor that relates the signal energy to the measurement energy. When $c = 1$, energy is preserved on average; when $c > 1$, measurements compress the signal energy; when $c < 1$, measurements amplify it.

Property 4 (Relation to condition number): The value of $\text{trace}(\Phi^\top \Phi)$ is related to the Frobenius norm of Φ : $\|\Phi\|_F^2 = \text{trace}(\Phi^\top \Phi)$. This connects c to the overall "energy" of the sensing operator.

A.8 Remarks

- 1) The theorem establishes a linear relationship between the expected reconstruction error in the signal space and the expected measurement error in the observation space, under the independent and standardized variable assumption.
- 2) The constant c is deterministic given a fixed sensing matrix Φ and can be computed explicitly as $c = n/\text{trace}(\Phi^\top \Phi)$.
- 3) This result is fundamental for understanding error propagation in compressed sensing systems and provides theoretical justification for the design and normalization of sensing matrices.
- 4) The assumption of independent and standardized variables is crucial. For correlated variables with covariance matrix $\Sigma = E[xx^\top]$, the relationship generalizes to:

$$E(\|x - E[x]\|_2^2) = \text{trace}(\Sigma), \quad E(\|y - E[y]\|_2^2) = \text{trace}(\Phi \Sigma \Phi^\top). \tag{S.18}$$
- 5) In practice, when the standardization assumption is not satisfied, one can apply z-score normalization: $\tilde{x}_i = (x_i - \mu_i)/\sigma_i$, where $\mu_i = E[x_i]$ and $\sigma_i = \sqrt{\text{Var}(x_i)}$, to make the theorem applicable. □

B. PROOF OF PROPOSITION 1

Proof. We provide a detailed proof of Proposition 1, showing that the ELACS recovery problem can be reformulated as a standard LASSO problem through appropriate variable transformations. The proof strictly follows the notation and definitions from the main text.

B.1 Problem Setup and Variable Transformation

Consider the original ELACS optimization problem from the main text:

$$\mathcal{F}(x) = \frac{1}{2} \|y - \Phi x\|_2^2 + (1 - \alpha) \|x\|_1 + \alpha \|x\|_2^2, \quad (\text{S.19})$$

where $\Phi \in \mathbb{R}^{m \times n}$ is the measurement matrix, $y \in \mathbb{R}^m$ is the observation vector, and $\alpha \in (0, 1)$ is a regularization parameter. Note that the original problem does not explicitly include the sparsity basis ω_b in the regularization terms; ω_b is introduced later in the transformation for sparse representation.

Define the scaled variable:

$$x^* = \sqrt{1 + \alpha} \cdot x. \quad (\text{S.20})$$

B.2 Transformation of the Data Fidelity Term

From equation (S.20), we have $x = \frac{x^*}{\sqrt{1 + \alpha}}$. We construct the augmented observation vector and measurement matrix as in Proposition 1:

$$y^* = \begin{bmatrix} y \\ 0 \end{bmatrix} \in \mathbb{R}^{(m+n)}, \quad \Theta = \frac{1}{\sqrt{1 + \alpha}} \begin{bmatrix} \Phi \\ \sqrt{\alpha} I \end{bmatrix} \in \mathbb{R}^{(m+n) \times n}, \quad (\text{S.21})$$

where $I \in \mathbb{R}^{n \times n}$ is the identity matrix and $0 \in \mathbb{R}^n$ is the zero vector.

B.3 Verification of Equivalence

We verify that:

$$\|y^* - \Theta x^*\|_2^2 = \|y - \Phi x\|_2^2 + \alpha \|x\|_2^2. \quad (\text{S.22})$$

Left-hand side:

$$\begin{aligned} \|y^* - \Theta x^*\|_2^2 &= \left\| \begin{bmatrix} y \\ 0 \end{bmatrix} - \frac{1}{\sqrt{1 + \alpha}} \begin{bmatrix} \Phi \\ \sqrt{\alpha} I \end{bmatrix} x^* \right\|_2^2 \\ &= \left\| \begin{bmatrix} y - \frac{1}{\sqrt{1 + \alpha}} \Phi x^* \\ -\frac{\sqrt{\alpha}}{\sqrt{1 + \alpha}} x^* \end{bmatrix} \right\|_2^2 \\ &= \left\| y - \frac{1}{\sqrt{1 + \alpha}} \Phi x^* \right\|_2^2 + \left\| \frac{\sqrt{\alpha}}{\sqrt{1 + \alpha}} x^* \right\|_2^2 \\ &= \left\| y - \Phi \frac{x^*}{\sqrt{1 + \alpha}} \right\|_2^2 + \frac{\alpha}{1 + \alpha} \|x^*\|_2^2. \end{aligned} \quad (\text{S.23})$$

Substituting $x^* = \sqrt{1 + \alpha} \cdot x$ from equation (S.20):

$$\begin{aligned} \|y^* - \Theta x^*\|_2^2 &= \|y - \Phi x\|_2^2 + \frac{\alpha}{1 + \alpha} \cdot (1 + \alpha) \|x\|_2^2 \\ &= \|y - \Phi x\|_2^2 + \alpha \|x\|_2^2. \end{aligned} \quad (\text{S.24})$$

B.4 Transformation of the Regularization Term

For the ℓ_1 regularization term, we have:

$$\begin{aligned} \|x\|_1 &= \left\| \frac{x^*}{\sqrt{1 + \alpha}} \right\|_1 \\ &= \frac{1}{\sqrt{1 + \alpha}} \|x^*\|_1. \end{aligned} \quad (\text{S.25})$$

B.5 Derivation of the Transformed Objective

Combining equations (S.24) and (S.25), the original objective function (S.19) becomes:

$$\begin{aligned} \mathcal{F}(x) &= \frac{1}{2} \|y - \Phi x\|_2^2 + (1 - \alpha) \|x\|_1 + \alpha \|x\|_2^2 \\ &= \frac{1}{2} (\|y - \Phi x\|_2^2 + \alpha \|x\|_2^2) + (1 - \alpha) \|x\|_1 \\ &= \frac{1}{2} \|y^* - \Theta x^*\|_2^2 + (1 - \alpha) \cdot \frac{1}{\sqrt{1 + \alpha}} \|x^*\|_1. \end{aligned} \quad (\text{S.26})$$

Define the regularization parameter as in Proposition 1:

$$\gamma = \frac{1 - \alpha}{\sqrt{1 + \alpha}}. \quad (\text{S.27})$$

The transformed objective function is:

$$\mathcal{F}^*(x^*) = \frac{1}{2} \|y^* - \Theta x^*\|_2^2 + \gamma \|x^*\|_1. \quad (\text{S.28})$$

B.6 Establishing the Equivalence

Since the transformation $x \mapsto x^* = \sqrt{1 + \alpha} \cdot x$ is bijective (one-to-one and onto), we have:

$$x = \frac{x^*}{\sqrt{1 + \alpha}}. \quad (\text{S.29})$$

For any minimizer $\hat{x}$ of $\mathcal{F}(x)$, the corresponding $\hat{x}^* = \sqrt{1 + \alpha} \cdot \hat{x}$ is a minimizer of $\mathcal{F}^*(x^*)$, and vice versa. Therefore:

$$\arg \min_{x \in \mathbb{R}^n} \mathcal{F}(x) = \frac{1}{\sqrt{1 + \alpha}} \arg \min_{x^* \in \mathbb{R}^n} \mathcal{F}^*(x^*). \quad (\text{S.30})$$

Equivalently, we obtain the key relationship from Proposition 1:

$$\arg \min_{x^* \in \mathbb{R}^n} \mathcal{F}^*(x^*) = \sqrt{1 + \alpha} \cdot \arg \min_{x \in \mathbb{R}^n} \mathcal{F}(x). \quad (\text{S.31})$$

Note that in the main text, Proposition 1 includes the DCT-based basis ω_b in the minimizer relationship. However, as shown above, the core equivalence does not require ω_b . The introduction of ω_b is for promoting sparsity in the transformed domain, as discussed below.

B.7 Incorporation of Sparsity Basis ω_b

In the main text, after Proposition 1, the solution is simplified using the DCT-based basis ω_b to promote sparsity. Specifically, the transformed objective is extended to:

$$\hat{x}^* = \arg \min_{x^* \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y^* - \Theta x^*\|_2^2 + \gamma \|\omega_b x^*\|_1 \right\},$$

where ω_b is an orthogonal matrix satisfying $\omega_b^\top \omega_b = I$. This extension is valid because: - The transformation $x^* \mapsto \omega_b x^*$ is a change of basis, and the ℓ_1 norm is invariant under

orthogonal transformations when applied to the coefficient vector. - The minimizer relationship becomes:

$$\arg \min_{x^* \in \mathbb{R}^n} \mathcal{F}^*(x^*) = \sqrt{1 + \alpha} \cdot \omega_b \cdot \arg \min_{x \in \mathbb{R}^n} \mathcal{F}(x),$$

which is consistent with the main text, provided that $\mathcal{F}(x)$ is defined without ω_b , but the solution in the transformed domain is sparse under ω_b .

B.8 Key Insights

This proof demonstrates that:

- 1) The elastic net regularization (combining ℓ_2 and ℓ_1 terms) in the original ELACS formulation can be transformed into a pure LASSO problem by absorbing the ℓ_2 penalty into the data fidelity term through system augmentation and variable scaling.
- 2) The equivalence enables the use of standard LASSO solvers and theoretical results for the ELACS problem.
- 3) The introduction of the sparsity basis ω_b is optional and can be applied post-transformation to promote sparsity in a specific domain.

B.9 Remarks on Generalization

While Proposition 1 in the main text mentions the DCT-based basis ω_b , the proof above shows that the core equivalence holds without ω_b . The result generalizes to any orthogonal basis Ψ (satisfying $\Psi^\top \Psi = I$) by replacing ω_b with Ψ in the transformed objective. The choice of DCT in this work is motivated by:

- 1) Empirical sparsity of natural images in the DCT domain.
- 2) Computational efficiency via fast DCT algorithms.
- 3) Well-established compression standards.

□